Alert Management Practice Report

1. Objective

The objective of this task is to design and implement an alert management workflow, including alert classification, prioritization, visualization, incident ticket creation, and escalation handling in a SOC environment.

2. Alert Classification System

An alert classification system was created using Google Sheets to map alerts with their corresponding MITRE ATT&CK techniques and priorities.

Sample Alert Classification Table

Alert ID	Type	Priority	MITRE Tactic
001	Phishing	High	T1566
002	SSH Brute-force	Medium	T1110
003	Malware Activity	Critical	T1059
004	Port Scan	Low	T1046

3. Mock Alert Testing

A mock alert was created:

- **Alert:** Phishing Email – Suspicious Link
- **Behavior:** User receives malicious email containing a phishing URL
- **Classification:** High Priority, MITRE T1566

This confirms the classification system works correctly.

Alert ID	Type	Priority	MITRE Tactic
001	Phishing	High	T1566

4. Alert Prioritization (Using CVSS)

Alerts were prioritized based on severity using CVSS scoring.

Example Prioritization

Alert Type	CVSS Score	Priority
Log4Shell Exploit	9.8	Critical
SSH Brute-force	6.5	Medium
Port Scan	3.7	Low

High CVSS score = Higher priority

Critical alerts require immediate response

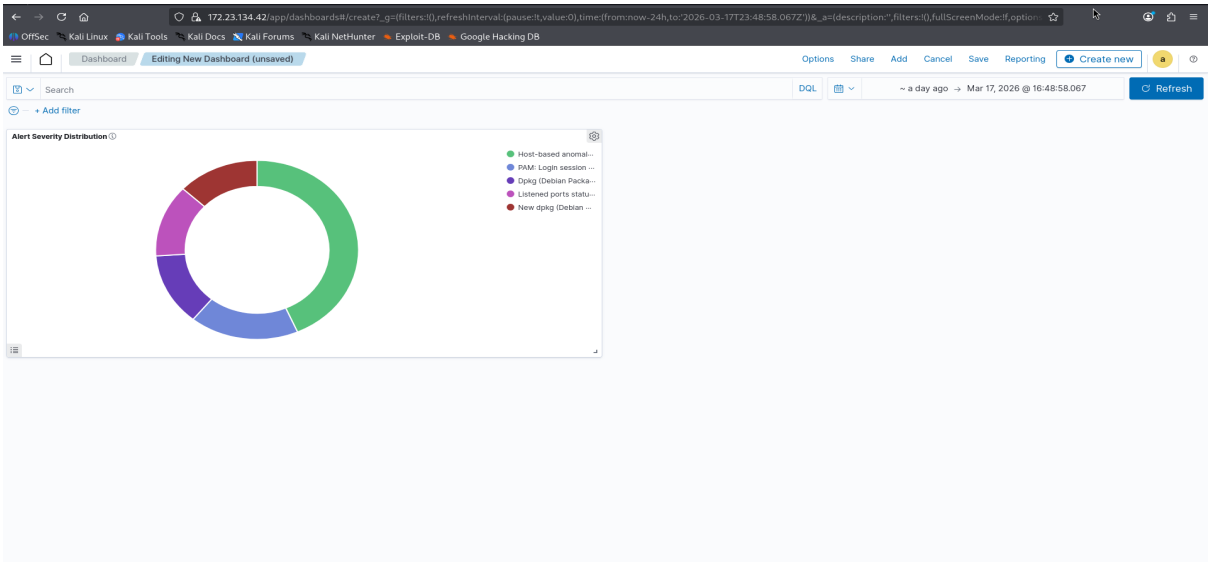
5. Dashboard Creation (Wazuh)

A dashboard was created in Wazuh to visualize alert severity levels.

Visualization Created

- Pie chart showing:
  - Critical alerts
  - High alerts
  - Medium alerts
  - Low alerts

This helps analysts quickly understand alert distribution.



6. Response Procedure (SOC Workflow)

The following response steps were defined:

1. Detect alert from SIEM
2. Classify alert based on type and severity
3. Prioritize using CVSS score
4. Investigate logs and indicators
5. Create incident ticket
6. Escalate if necessary
7. Take remediation action

7. Escalation Email

Subject: Critical Alert Escalation – Ransomware Detected

Dear Tier 2 Team,

A critical alert has been identified involving suspected ransomware activity on Server-X. Indicators include the file *crypto_locker.exe* and suspicious communication from IP address 192.168.1.50. The alert has been classified as Critical based on CVSS scoring and potential impact. Immediate investigation is recommended to prevent further spread and data compromise. Initial triage confirms malicious behavior consistent with ransomware patterns.

Please prioritize this incident and proceed with deeper analysis and containment.

Regards,
SOC Analyst L1